class-5

#1 The reactance of a coil equals the resistance of a 10 kΩ resistor at a frequency of 5 kHz. Determine the inductance of the coil.

2) Determine the capacitance required to establish a capacitive reactance that will match that of a $\boxed{2\,mH}$ coil at a frequency of 50 kHz.

Ans

1)

$$\boxed{X_L = \omega L}$$

$$X_L = 2\pi f L$$

$$L = \frac{X_L}{2\pi f} = \frac{10\times10^3}{2\pi\times5\times10^3} = 0.32\ H.$$

2)

$$X_C = X_L$$

$$\Rightarrow \frac{1}{\omega C} = \omega L$$

$$\Rightarrow C = \frac{1}{(2\pi f)^2 L} = \frac{1}{(2\pi\times50\times10^3)^2\times2\times10^{-3}}$$

$$= 5\,nF$$

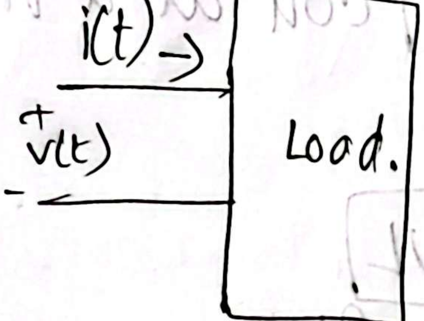
Instantaneous and Average Power

#. The instantaneous power (in watts) is the power at any instant of time.

#. The average power (in watts) is the average of the instantaneous power over one period.

$v(t) = V_m \cos(\omega t + \theta_v)$

$i(t) = I_m \cos(\omega t + \theta_i)$



$$p(t) = v(t)\, i(t)$$

$$= V_m \cos(\underbrace{\omega t + \theta_v}_{A}) \cdot I_m \cos(\underbrace{\omega t + \theta_i}_{B})$$

$$\rightarrow \cos A \cdot \cos B = \frac{1}{2}\left[\cos(A-B) + \cos(A+B)\right]$$

$$p(t) = \underbrace{\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)}_{\text{Time independent}} + \underbrace{\frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)}_{\text{Time dependent}}$$

instantaneous ~~value~~ power.

Avg. value

Time independent

Time dependent.

$$P_{av} = \frac{1}{T}\int_0^T p(t)\,dt$$

$$= \frac{1}{T}\int_0^T \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)\,dt + \frac{1}{T}\int_0^T \cos(2\omega t + \theta_v + \theta_i)\,dt$$

$$= \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)$$

$$P_{av} = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)$$

$$= \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}}\cos(\theta_v - \theta_i)$$

$$= V_{rms} \cdot I_{rms}\cos(\theta_v - \theta_i)$$

$$= V_{rms} \cdot I_{rms}\cos(\theta) \quad [\theta = \theta_v - \theta_i]$$

Q) Find the instantaneous power and the avg. power absorbed by the passive linear network.

a) $v(t) = 120\cos(377t + 45^\circ)$ volt.

$i(t) = 10\cos(377t - 10^\circ)$ amp.

Ans

$v(t) = V_m\cos(\omega t + \theta_v)$

$i(t) = I_m\cos(\omega t + \theta_i)$

$$P_{av} = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)$$

$V_m = 120$ $I_m = 10$

$\theta_v = 45^\circ$ $\theta_i = -10^\circ$

$$P_{av} = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i)$$

$$= \frac{120 \times 10}{2} \cos(45^\circ - (-10^\circ))$$

$$= 344.15 \text{ W}$$ Ans

(b) Instantaneous power

$$P(t) = v(t)\, i(t)$$

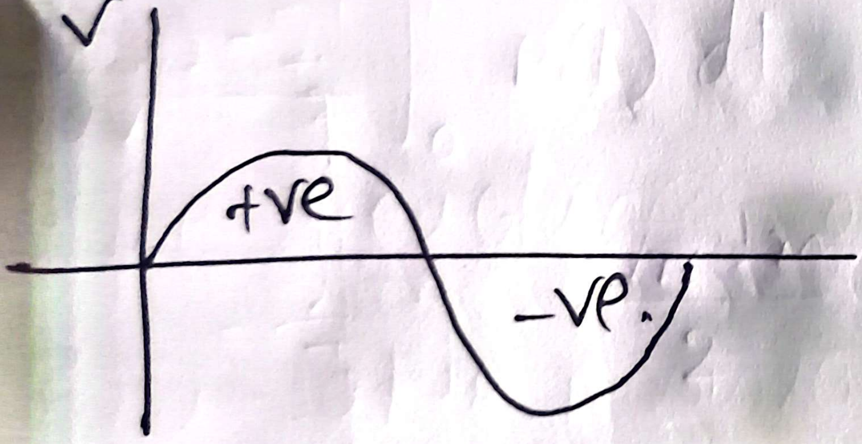
$$= 120 \cos(377 + 45^\circ) . 10 \cos(377t - 10^\circ)$$

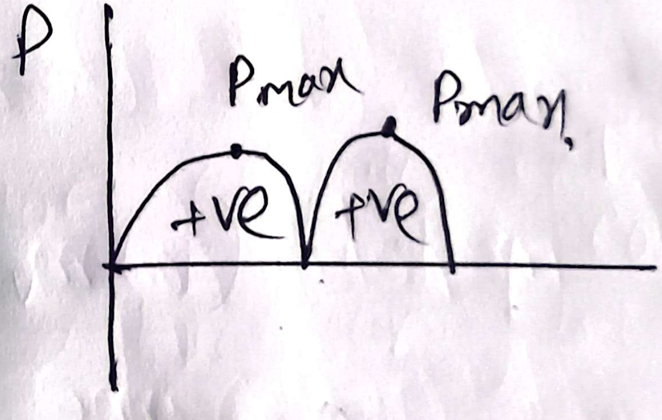
#. $\cos A . \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$

$$= 120 \times 10 \times \frac{1}{2} \times \cos(55^\circ) + 120 \times 10 \times \frac{1}{2} \times \cos(2 \times 377t + 45^\circ - 10^\circ)$$

$$= 344.15 + 600 \cos(754t + 35^\circ) \text{ watt}$$

Ans





for
power.